Performance Analysis of Apache ZooKeeper's Write Pipeline

Software Performance Engineering — Project Stage II

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Abstract. ZooKeeper serves reads at any server, but every write goes through one leader and a quorum. We measured the cost of this write path on Google Compute Engine with a 2^{6-1} factorial experiment, Universal Scalability Law curves at three write ratios and five eBPF probes. The share of writes and the number of clients explain most of the variation, and turning `fsync` off helps at light load but hurts at saturation. With writes the curves show contention but no crosstalk, and the CPU the leader spends per operation near saturation matches the bottleneck demand the fit implies. We also found that a logging client had distorted our Stage I results.

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1 Introduction

Apache ZooKeeper is a coordination service used by Kafka, HBase and Hadoop for leader election, configuration and locks [1, 2]. Its replication protocol, Zab [3], makes reads and writes behave very differently: a read is answered by the server the client is connected to, while a write goes to the leader, is logged and acknowledged by a majority and only then committed. Reads should therefore scale with the number of servers and writes should not, and the write pipeline is where we look for the limits of scalability. We chose ZooKeeper because this single-leader design is shared by many coordination and consensus systems, and because it exposes per-stage timers that let us check a model against the pipeline itself.

Stage I measured one laptop with Docker and fitted the Universal Scalability Law (USL) [4, 5]. For Stage II we first found a client-side artifact in that data (Section 3). On dedicated virtual machines we then ran a fractional factorial experiment (Sections 5–6.1), fitted USL curves at 0, 30 and 50% writes with ZooKeeper's per-stage timers (Section 6.2), and looked inside the leader with eBPF (Section 6.3).

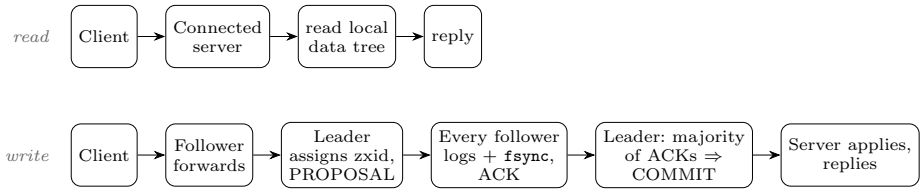


Fig. 1. Read and write paths in ZooKeeper.

2 System Description

ZooKeeper keeps a tree of small data nodes (znodes) in memory. Clients read and write whole znodes of at most 1 MB, can set one-time *watches* that notify them of a change, and can create *ephemeral* znodes that disappear when their session ends. Each client holds a session with one server, kept alive by heartbeats. Every change is appended to a transaction log, and the in-memory tree is written to a snapshot from time to time. With $N = 2f + 1$ voting servers, the group elects a *leader*, survives f failures and stops accepting writes when no majority is reachable [3]. *Observers* also receive the committed writes, but they do not vote. Writes are linearizable, while reads are only sequentially consistent per session and may be slightly stale. That relaxation is what lets a follower answer a read alone [1]. Figure 1 shows both paths.

Inside the leader, a write passes a chain of processors. Request preparation, a single thread, validates it and assigns its zxid. The log-sync stage, also a single thread, appends it to the log and calls `fsync`, and it writes every request queued at that moment in one batch, which is called group commit. Once a majority has acknowledged, the commit stage applies the write. The followers run the same log-sync stage before they acknowledge. Reads go through the commit stage too, and a read waits there while a write of the same session is pending. A read thus costs one hop, while a write needs a forward, a proposal, a log write and `fsync` on every follower, the acknowledgments and a commit.

3 Stage I

In Stage I the same configuration (3 servers, 20 clients, 30% writes) gave 6,364 ops/s in the sweep and 14,832 ops/s when we repeated it later. The jar used for the sweep bundled Logback, and its default configuration logs the ZooKeeper client at DEBUG level. It printed about 145,000 lines in a 15 s run, more than one per operation, and every client thread ended up waiting on the console. A one-line Logback configuration fixes this, and the runner now aborts any run in which the client prints more than a few lines. Those fits had given the same contention coefficient (≈ 0.23) for 1, 3 and 5 servers, and we had concluded that the contention was on the server side but not caused by Zab. A bottleneck in the client gives exactly that picture, because it does not depend on the number of servers. With the quiet client the coefficient became 0.025, 0.088 and 0.112

for 1, 3 and 5 servers. On the laptop all servers shared one CPU, so every extra server added load to the same machine, which is not the case with separate cloud machines. Repetitions of one variant on the laptop also differed by up to a factor of two. For these reasons Stage II uses separate machines for client and servers, replicates and a randomized order. The Stage I experiments we did not repeat (YCSB, leader failover, injected network delay) are summarized in Appendix A as context only.

4 Testbed and Methodology

Deployment. All Stage II runs use Google Compute Engine in one zone (europe-west1-b). The load generator has its own n2-standard-8 machine (8 vCPUs, 32 GB), and every ZooKeeper server has an n2-standard-2 (2 vCPUs, 8 GB). Each server has two extra 20 GB disks, one pd-balanced and one pd-ssd, formatted with ext4. The transaction log goes on one of them according to the factor *log disk*, while snapshots stay on the boot disk. Before each block of runs the servers are stopped, their data is erased and they are restarted with the new configuration. Because a wrong setting would silently spoil a whole block, the runner then reads the `forceSync` flag and the log directory back from every server before the first run.

Load generator. A Java client opens one ZooKeeper session per simulated client, and the connect string spreads the sessions over all servers. Each client runs a closed loop of synchronous calls with no think time. In every iteration it picks one of 200 znodes at random and sends a `setData` with probability equal to the write ratio, or a `getData` otherwise. Operations that start in the first 5 s are executed but not recorded, which keeps JIT compilation and cold caches out of the measurement, and every series starts with a discarded run. After that, a factorial run measures 20 s and a USL run 30 s. Even the slowest factorial run completed more than 30,000 writes, so its 99th percentile rests on over 300 samples, and since all latencies are kept, the percentiles are exact. The client's CPU stayed below 47% in every factorial run and below 28% in the USL runs with writes. Before the experiments we set 70% as the level above which we would discard a run. This is a rule of thumb rather than a derived value, since queueing inside the client grows as $1/(1 - U)$ at any load and a busier client always says less about the servers.

ZooKeeper's per-stage timers. The `mntr` command reports a count and a sum per stage (queue times, `fsync` time, batch size, time to quorum acknowledgment, commit to applied). Snapshots of every server before and after a run give the mean of that run as the ratio of the differences. The `fsync` timer truncates each call to whole milliseconds, so the `fsync` is also measured with eBPF.

Threats to validity. The virtual machines share physical hosts with other tenants. Replicates and the random order spread that noise over all treatments, but they

Table 1. Fixed parameters. Defaults are those of ZooKeeper 3.9 [6].

Parameter	Value
System	
Server machines	n2-standard-2: 2 vCPUs (N2 series, Intel Xeon), 8 GB, Debian 12, kernel 6.1
Network	one VPC subnet in one zone; up to 10 Gbps egress per VM (GCE default for this type)
Server JVM; versions	JDK 17, 1 GB heap, G1; ZooKeeper 3.9.3, client library 3.9.0
Observers; `leaderServes`; snapshots	0; yes (default); boot disk (pd-standard)
`snapCount`, `preAllocSize`, `maxBatchSize`, `flushDelay`	100,000, 64 MB, 1000, 0 (defaults)
`globalOutstandingLimit`; `maxClientCnxns`; `tickTime`	1000 (default); no limit; 2000 ms
Workload	
Operations; key space	`getData/setData` only; 200 znodes, uniform, new pool per run
Client model; session placement	closed loop, synchronous, no think time; random server
Load generator	n2-standard-8 (8 vCPUs, 32 GB), one JVM, one thread per client
Run length	5 s warm-up + 20 s (factorial) or 30 s (USL) measured

do not remove it, and two replicates of the same factorial run differ by 11% at the median and by up to 48%. Sessions are placed on a random server, so the leader's share of the sessions changes from run to run. In the extra USL curves, where we recorded it, the share averaged a third but ranged from 19% to 44% at 8–16 clients. Finally, `forceSync=no` is unsafe in production, and we use it only as a probe.

5 Experimental Design

Parameters and metrics. Table 1 lists the fixed parameters and Table 2 the factors. The primary response is throughput (successful operations per second), the quantity the USL models. The secondary response is the 99th-percentile write latency, because a coordination service sits on the critical path of other systems. We do not use the mean latency as a separate response: in a closed loop without think time Little's law gives the mean response time $R = N/X$, so it carries no information beyond throughput. Read latency is recorded and reported, but reads do not enter the write pipeline. Failed operations (there were none) and the client's CPU are recorded as validity checks.

Factors and levels. Each factor tests one guess about where a write spends its time (Table 2). The two client levels, 10 and 160, come from the Stage I curve. On the new testbed 10 clients already run at 74% efficiency (Section 6.2), so in practice the levels are moderate load and deep saturation. For the write ratio, Hunt et al. give 2:1 to 100:1 read-to-write ratios as ZooKeeper's target workloads [1]. A write ratio of 0.1 lies inside that range, and 0.5 loads the write path well beyond it.

Design. We use the half fraction 2^{6-1} with $F = ABCDE$ (defining relation $I = ABCDEF$) [7]. It has resolution VI: main effects are aliased with five-factor interactions and two-factor interactions with four-factor ones, so all six

Table 2. Factors and levels. "Hard": changing it needs a rebuild of the servers.

Factor	−1	+1	Hard	Why
A number of servers	3	5	yes	quorum size: proposals and ACKs per write
B write ratio	0.1	0.5		share of requests that take the write path
C clients	10	160		light load vs well into saturation
D forceSync	yes	no	yes	removes the fsync from the write path
E payload (bytes)	128	4096		bytes logged, replicated and synced per write
F log disk	pd-balanced	pd-ssd	yes	device under the fsync

Table 3. Sign table of the 2^{6-1} design with the mean of $\log_{10}$ throughput over the two replicates; the last row is the effect q_j of each factor and the grand mean q_0.

Run	A	B	C	D	E	F	$\overline{\log_{10} y}$	Run	A	B	C	D	E	F	$\overline{\log_{10} y}$
1	−	−	−	−	−	−	4.409	17	−	−	−	−	+	+	4.226
2	+	−	−	−	−	+	4.402	18	+	−	−	−	+	−	4.227
3	−	+	−	−	−	+	3.944	19	−	+	−	−	+	−	3.752
4	+	+	−	−	−	−	3.986	20	+	+	−	−	+	+	3.796
5	−	−	+	−	−	+	4.723	21	−	−	+	−	+	−	4.656
6	+	−	+	−	−	−	4.902	22	+	−	+	−	+	+	4.731
7	−	+	+	−	−	−	4.300	23	−	+	+	−	+	+	4.183
8	+	+	+	−	−	+	4.306	24	+	+	+	−	+	−	4.215
9	−	−	−	+	−	+	4.472	25	−	−	−	+	+	−	4.343
10	+	−	−	+	−	−	4.511	26	+	−	−	+	+	+	4.415
11	−	+	−	+	−	−	4.093	27	−	+	−	+	+	+	3.980
12	+	+	−	+	−	+	4.077	28	+	+	−	+	+	−	3.984
13	−	−	+	+	−	−	4.684	29	−	−	+	+	+	+	4.575
14	+	−	+	+	−	+	4.775	30	+	−	+	+	+	−	4.597
15	−	+	+	+	−	+	4.106	31	−	+	+	+	+	−	4.082
16	+	+	+	+	−	−	4.360	32	+	+	+	+	+	+	4.177
q	+0.029	-0.228	+0.149	+0.015	-0.066	-0.007	4.312								

main effects and fifteen two-factor interactions are estimated from 32 runs per replicate if interactions of order four and higher are negligible. A quarter fraction would have resolution IV and alias two-factor interactions in pairs, which is exactly what three of our hypotheses (H2–H4, listed below) are about. Table 3 is the sign table: the full 2^5 table of A–E with F as the product of the other five columns. Two replicates give 64 runs and 32 degrees of freedom for error, which is what the credit allowed. With the measured error this gives intervals of ± 0.015 on each $\log_{10}$ effect, so a change of about 7% between levels is detectable.

Restricted randomization. A, D and F need a rebuild of the servers, which takes several minutes. Fully randomizing them would have meant 64 rebuilds, so we grouped the runs that share their levels into blocks of four (eight blocks per replicate, 16 rebuilds in total). The block order is randomized in each replicate, the run order inside each block as well, and the plan was generated with a fixed seed before the first run. The whole-plot contrast ADF is aliased with the subplot interaction BCE, so BCE is confounded with the blocks. Grouping runs this way makes the experiment a split-plot design. A, D and F change only from one block to the next, so for them a block, not a run, is one independent observation, and testing them against the run-to-run error alone would overstate their precision. We therefore test all effects against the pooled run error and, in addition, the

seven whole-plot contrasts (A, D, F, AD, AF, DF, ADF) against a whole-plot error estimated from the 16 block means (replicate × treatment, $(2-1)(8-1) = 7$ degrees of freedom).

Analysis. With s_{ij} the sign of run i for effect j, $\bar{y}_i$ the mean of the $r = 2$ replicates and $n = 32$, the effect is $q_j = \frac{1}{n}\sum_i s_{ij}\bar{y}_i$, with $SS_j = nrq_j^2$, $s_e^2 = \sum_{i,k}(y_{ik} - \bar{y}_i)^2/(n(r-1))$ and the interval $q_j \pm t_{[0.975;\, n(r-1)]}\, s_e/\sqrt{nr}$. Effect j explains $100\, SS_j/SST$ percent of the variation. Since 31 effects tested at 95% would give about two false positives by chance alone, we rank effects by their share of the variation instead of relying only on significance. When $y_{\max}/y_{\min} > 10$ we analyse $\log_{10} y$ (residual plots in Appendix F). Before using the analysis scripts on the measurements, we ran them on synthetic data with known effects.

Hypotheses. From the structure of the write pipeline (Section 2) and the quiet-client Stage I runs we expected: **H1**: turning `fsync` off (D) raises throughput and lowers the write p99. **H2**: that benefit grows with the write ratio (B×D), since reads never touch the log. **H3**: a faster log disk (F) helps only while `fsync` is on (D×F). **H4**: five servers (A) lower throughput and raise the write p99, mostly through writes (A×B). **H5**: the number of clients (C) has the largest main effect on throughput.

6 Results

6.1 Factorial experiment

All 64 runs completed without errors. Throughput ranged from 5,335 to 81,088 ops/s and the write p99 from 1.2 to 61 ms. Both ratios exceed 10, and the residuals of the additive model widen with the predicted value, so both responses are analysed on $\log_{10}$, where the residuals look homogeneous and roughly normal (Fig. 6). Effects are then multiplicative factors (Table 4); all shares of variation below are of the $\log_{10}$ response. The error explains 2.1% (throughput) and 1.3% (p99), small next to factors that change throughput up to fifteen-fold even though replicates differ by about 11%.

Throughput. The write ratio (B) has the largest effect: going from 10% to 50% writes divides throughput by 2.9 (60% of the variation). If a read costs the servers r and a write w, a throughput ratio ρ gives $(0.9r + 0.1w)/(0.5r + 0.5w) = \rho$. This holds only when server capacity sets throughput, that is, at 160 clients, where $\rho = 0.32$ and $w \approx 12r$: a write costs the system about twelve reads, an aggregate figure because reads are spread over all servers while writes load the leader most. The number of clients (C) comes second (×1.98, 26%), and 4 KiB payloads (E) cost 26% of throughput (5%).

The largest interaction is clients × `forceSync` (C×D, 3.6%, Fig. 2). With 10 clients, turning `fsync` off raises throughput from 12,381 to 17,150 ops/s (geometric means, +39%), but with 160 clients it lowers it from 31,776 to 26,270 ops/s (−17%). This reversal holds in every cell, and at 160 clients the ratio without

Table 4. Largest effects ($\log_{10}$ model; all 31 in Appendix B): q, the factor (×) from the low to the high level, the 95% interval of q and the share of the variation (%). †: interval includes zero.

	throughput (ops/s), $\log_{10}$			write p99 (ms), $\log_{10}$		
Effect	q (×)	95% CI	%	q (×)	95% CI	%
C	0.149 (×1.98)	[0.133, 0.164]	25.6	0.534 (×11.68)	[0.511, 0.556]	93.7
B	-0.228 (×0.35)	[-0.244, -0.213]	60.3	0.041 (×1.21)	[0.018, 0.064]	0.6
E	-0.066 (×0.74)	[-0.081, -0.051]	5.0	0.061 (×1.32)	[0.038, 0.084]	1.2
C×D	-0.056 (×0.77)	[-0.071, -0.041]	3.6	0.059 (×1.31)	[0.036, 0.081]	1.1
A	0.029 (×1.14)	[0.014, 0.044]	1.0	0.021 (×1.10)†	[-0.001, 0.044]	0.2
D	0.015 (×1.07)†	[-0.001, 0.030]	0.3	-0.042 (×0.82)	[-0.065, -0.019]	0.6
E×F	0.021 (×1.10)	[0.006, 0.036]	0.5	-0.004 (×0.98)†	[-0.027, 0.018]	0.0
A×B	0.000 (×1.00)†	[-0.016, 0.015]	0.0	-0.034 (×0.85)	[-0.057, -0.012]	0.4
A×C	0.018 (×1.09)	[0.003, 0.033]	0.4	0.010 (×1.05)†	[-0.013, 0.033]	0.0
B×C	-0.016 (×0.93)	[-0.032, -0.001]	0.3	-0.020 (×0.91)†	[-0.043, 0.003]	0.1
B×E	0.003 (×1.01)†	[-0.012, 0.019]	0.0	-0.022 (×0.90)†	[-0.045, 0.000]	0.2

and with `fsync` is between 0.79 and 0.87 at all levels of the other factors. At 10 clients each write waits for its own `fsync`, so removing it shortens every request. At 160 clients group commit already spreads one `fsync` over many requests, so removing it saves little, and our data do not show why it then costs throughput. The probes ran at 80 clients, where the reversal does not appear. There, without `fsync`, the log-sync stage flushes 1.9 requests at a time instead of 16–17, yet the leader spends the same CPU per operation (59 against 60 μs). Explaining the reversal would need probes at 160 clients.

Five servers raise throughput by 14% (×1.14, 1% of the variation), and the effect is significant against both errors (×1.08–1.21 in the whole-plot test). The gain grows with the load (A×C), from ×1.05 at 10 clients to ×1.24 at 160. It is the same at both write ratios (A×B = ×1.00), so it cannot come from the extra servers answering more reads. One possible reason is the leader's share of the sessions, which we did not record in the factorial runs. Sessions are spread over all servers, so the leader serves a third of them with three servers and a fifth with five, and client connections take 0.4 of the leader's two cores (Section 6.3). With fewer sessions the leader keeps more CPU for coordination, which only matters once it is saturated. The log disk (F) has no main effect under either error and appears only in a small E×F interaction (0.5%).

Write tail latency. The write p99 depends mostly on load. Going to 160 clients multiplies it by 11.7 (94% of the variation), from 2.7 to 31.7 ms in geometric means. With sixteen times more clients and less than twice the throughput, each request simply finds a longer queue at the leader. No other effect changes the p99 by more than a third. Payload gives ×1.32, the write ratio ×1.21, `forceSync` ×0.82 and five servers ×1.10 (significant only in the whole-plot test). Reads stay fast throughout, with a median of 0.27 ms at 10 clients and 0.68 ms at 160.

Pooled against whole-plot error. The whole-plot intervals are narrower than the pooled ones, so rebuilding the servers added no detectable variance. None of the

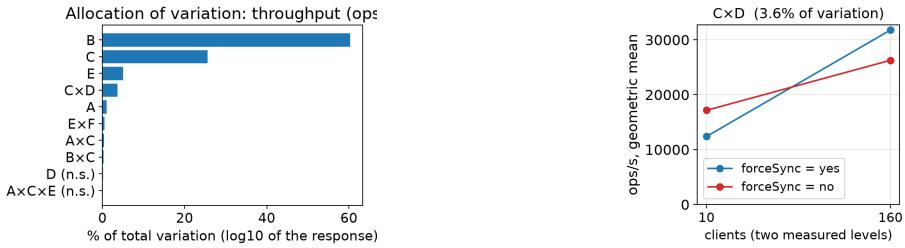


Fig. 2. Left: share of the variation of $\log_{10}$ throughput per effect (ten largest). Right: clients × `forceSync`; the lines only join the two measured levels.

four whole-plot interactions is detectable, D×F included. With only 7 degrees of freedom, we call an effect that is significant in just one of the two tests borderline.

Hypotheses. **H1** holds only at 10 clients. The main effect of D on throughput (×1.07) is borderline and much smaller than C×D, which changes its sign, and D lowers the write p99 by 18%. **H2** is rejected (B×D, ×1.04, not significant). **H3** is rejected too, because neither F nor D×F is detectable. The two disks turned out to have almost the same latency (Section 6.3), so the levels were too close to test it. **H4** is rejected for throughput, since five servers are faster, and holds in part for the p99, which five servers raise with 10% writes but not with 50% (A×B, ×0.85). **H5** is rejected for throughput, where the write ratio matters more, and holds for the write p99.

6.2 USL curves

We measured one configuration (3 servers, 128 B, `forceSync` on, pd-ssd) at three write ratios: 0, 30 and 50%, each at 13 client counts from 1 to 160 with three replicates per level in random order. In the notation of the course,

$$X(N) = \frac{\lambda N}{1 + \sigma(N-1) + \kappa N(N-1)}, \tag{1}$$

where λ is the throughput of one client, σ the contention (serial) coefficient and κ the crosstalk (coherency) coefficient. The fit minimizes the relative error, so that a 3% miss counts the same at 600 and at 6,000 ops/s. With only three replicates per level, a formula for the intervals would rest on strong assumptions, so we use a bootstrap instead. We resample the replicates of each load level, refit, repeat this 2,000 times and keep the middle 95% of the refitted values. To decide whether κ is needed, AIC and an F-test check whether it lowers the error by more than an extra parameter would by chance. If it does not, we keep the contention-only (Amdahl) model. We validate each fit by fitting $N \leq 64$ and predicting the three highest loads. The timers and probes below were recorded along the 30% curve.

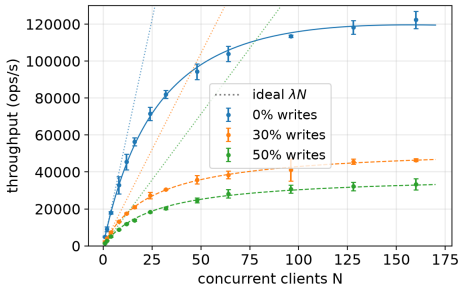


Fig. 3. Throughput against clients at three write ratios (mean of three replicates with its 95% confidence interval) with the chosen fit (USL at 0% writes, Amdahl at 30 and 50%) and the ideal linear scaling λN (dotted).

Table 5. Fits of the three curves. For 30 and 50% writes κ is not supported and the Amdahl fit is shown; intervals are 95% bootstrap intervals.

Writes	λ (ops/s)	σ	κ	F-test p	$X(160)$
0%	4891 [4782, 4993]	0.029 [0.027, 0.030]	4.0e-5 [3.2e-5, 4.7e-5]	0.00	122321
30%	2088 [2018, 2169]	0.039 [0.037, 0.041]	not supported	0.72	46436
50%	1426 [1387, 1459]	0.037 [0.036, 0.039]	not supported	1.00	33418

30% writes. Throughput rises from 2,165 ops/s with one client to 46,436 ops/s with 160 (Fig. 3). The full USL fit gives $\lambda = 2{,}082$ ops/s (95% interval 1,996–2,169) and $\sigma = 0.0385$ (0.0345–0.0414). The crosstalk term is not supported, since the Amdahl model has the lower AIC (-241.4 against -239.5), the F-test gives $p = 0.72$ and 37% of the bootstrap fits put κ at zero. We therefore keep the Amdahl fit ($\lambda = 2{,}088$, $\sigma = 0.039$, Table 5), which bounds throughput by $\lambda/\sigma \approx 53{,}500$ ops/s. At 160 clients the system delivers 87% of that bound, and the efficiency $X(N)/(N\lambda)$ has fallen to 0.14, from 0.74 at 10 clients. Fitted on $N \leq 64$ only, the model predicts 96, 128 and 160 clients within 1.9–5.1%.

50% writes. Here λ drops to 1,426 ops/s while σ stays at 0.037 (0.036–0.039). Again κ is not supported ($p = 1.0$), and the hold-out error is 4.1%. A bootstrap of the difference between the two Amdahl fits gives $\sigma_{50} - \sigma_{30} = -0.002$ with a 95% interval of $[-0.005, +0.001]$, so we detect no change. More writes make every request more expensive but leave the serialized share of the work as it was, and the ceiling λ/σ falls from 53,500 to 38,200 ops/s. The curve keeps rising up to 33,418 ops/s at 160 clients, without a retrograde region.

0% writes. Reads alone reach 122,321 ops/s, with $\lambda = 4{,}891$ and $\sigma = 0.029$. Here κ is supported ($p < 0.001$), with the peak $N_{\max} \approx 157$ at the edge of the range. But the curve has not turned down, and from 64 clients on the client was 59–70% busy, near our 70% limit, so we claim no retrograde region. Reads bypass the leader, so σ here is each server's own contention, and with the client near its limit κ may come from the client, as in Stage I.

Response time. Combining the Amdahl fit with Little's law for a closed loop without think time gives

$$R(N) = \frac{N}{X(N)} = \frac{1 + \sigma(N-1)}{\lambda}, \tag{2}$$

a floor of $1/\lambda = 0.48\,\text{ms}$ plus $\sigma/\lambda = 19\,\mu\text{s}$ for every added client. Once the system saturates, each new client only adds waiting time. Because $R(N)$ is derived from the throughput fit, its match with the measured mean N/X (within 3.7%, Appendix D) follows by construction, and the measured median and 95th percentile lie on either side of it.

Per-stage times. ZooKeeper's timers on the leader along the 30% curve are in Appendix E (Table 13, Fig. 5). The intervals overlap and do not add up: the time to quorum acknowledgment starts when the proposal is created and includes the leader's own log sync and the followers' round trip. From 1 to 160 clients the server-side write latency grows from 0.87 to 5.57 ms. The time to quorum acknowledgment grows most in absolute terms (+2.07 ms), followed by the queue in front of request preparation (+1.35 ms, most of it beyond 64 clients) and the time from commit to applied (+1.11 ms). The log-sync queue levels off at about 0.7 ms, because the batch grows from 1 to 17.6 requests per `fsync` (18.5 in the separate probe runs). The `fsync` timer truncates to whole milliseconds, hence its 0.06 ms at $N = 1$. A read served by the leader takes 2.79 ms at 160 clients against 0.97 ms on average, 1.48 ms of it in the commit stage.

6.3 eBPF probes

Four probes are our own bpftrace programs [8] and one is bcc's `runqlat`. They measure `fsync` latency (syscall tracepoints), block-I/O latency per disk (block tracepoints, since bcc's `biolatency` kprobe is not traceable on kernel 6.1), run-queue latency, TCP retransmissions, and CPU per thread (a 99 Hz `profile` probe keyed by the stage's thread name). They ran on the leader in two sets of separate runs: along the 30% curve at 1, 16, 64 and 160 clients (Table 6), and at 80 clients and 50% writes at both levels of `forceSync` and log disk (Appendix C), each probed run placed between two identical unprobed runs to measure the overhead.

The `fsync` and the disk. With `forceSync=no` the probe sees no `fsync` at all, which confirms that factor D was applied. Along the curve an `fsync` grows from 0.55 ms at one client to 1.1 ms at 160, as the batch grows from 1 to 18.5 requests. At 80 clients it takes 1.0 ms on average on both disks, and its CDF (Fig. 4) has two steps, with 21–27% of the calls at 256–512 μs and 68–75% at 1–2 ms. The block-I/O CDFs of the two disks almost coincide, with 85–88% of the requests in the 256–512 μs bucket. Neither disk is faster, which agrees with F having no effect, and a separate `fio` test (4 KiB writes, an `fsync` after each) confirms it with 1.31 ms per `fsync` on pd-balanced and 1.33 ms on pd-ssd. The

Table 6. Probes along the 30% curve: throughput, ZooKeeper's counters and the leader's CPU per thread.

clients	$N = 1$	$N = 16$	$N = 64$	$N = 160$
ops/s with probes	1840	18958	37239	44450
ops/s without (mean of 2)	1726	19746	35836	42842
fsync calls (eBPF)	15719	32824	23716	20016
fsync mean, eBPF (μs)	551	724	1015	1104
fsync mean, ZooKeeper (ms)	0.06	0.27	0.70	0.76
requests per flush	1.0	4.8	13.1	18.5
to quorum ACK (ms)	0.77	1.40	2.26	2.95
CPU of the leader (cores of 2)				
replication to followers	0.07	0.42	0.52	0.58
client connections (I/O)	0.18	0.29	0.36	0.43
request preparation	0.05	0.17	0.23	0.25
commit	0.04	0.11	0.15	0.17
log sync	0.05	0.11	0.10	0.11
commit workers	0.02	0.04	0.08	0.09
whole JVM	0.52	1.22	1.51	1.70

2.7 block requests per `fsync` suggest that ext4 also commits its journal, which would explain the 1–2 ms step. Without `fsync` the log disk sees only about 390 large background write-backs in 45 s, which no request waits for.

CPU, scheduler and network. The leader's JVM goes from 0.52 of its two vCPUs at one client to 1.70 at 160, mostly replication (0.58) and client connections (0.43); request preparation uses 0.25 and log sync 0.11. The run-queue p99 grows from 63 to 511 μs, garbage collection stays under 0.003 cores, and no packet is retransmitted.

Probe overhead. Probed runs differ from their unprobed neighbours by -4% to $+7\%$, within the neighbours' own spread (up to 17%; 43% in the 80-client set, used only for counts, distributions and CPU shares).

7 Discussion: What Limits Scalability

Log sync. Table 7 maps the USL terms to the pipeline. The log-sync stage is a single thread, so it is the first candidate for σ. At 160 clients the leader logs about 13,300 writes per second in batches of 18.5, which makes 720 `fsync`s per second of 1.1 ms each. The thread is in `fsync` about 80% of the time (86–87% at 16 and 64 clients), but it does not cap throughput: more requests make the batch larger and the cost per request smaller, which is why neither D nor F helps at saturation.

Leader CPU. No other thread comes close to a full core, which leaves the leader's CPU as a whole. The Amdahl fit caps throughput at $\lambda/\sigma \approx 53{,}500$ ops/s, so the bottleneck needs $D = \sigma/\lambda \approx 18.7\,\mu$s per operation. If that bottleneck is the leader's two vCPUs ($U = XD/2$), each operation needs 37 μs of CPU. The thread probe measures this directly along the curve: 40.4 μs at 64 clients and

Table 7. USL terms and number of servers: candidate stages and evidence.

Term	Candidate stages	Evidence
σ	The leader's CPU: replication, client connections, ordering, log sync	$\sigma \approx 0.038$ at 30 and 50% writes; demand $2\sigma/\lambda = 37\,\mu$s/op against 38–40 measured; JVM at 1.5–1.7 of 2 cores; no thread near a full core; D and F do not help at saturation
κ	Interference between pairs of clients (sessions, GC, commit waits)	Not supported with writes ($p = 0.72$, 1.0); GC under 0.003 cores; supported at 0% writes, possibly from the client
λ	One request alone: round trips, serialization, payload, `fsync`	0.48 ms per request; 4 KiB payloads $\times 0.74$; quorum ACK 0.72 ms at $N = 1$ (0.77 in the probe runs)
Servers (A)	More followers per proposal vs. fewer sessions on the leader	$\times 1.14$, mostly at 160 clients; same at both write ratios; no retransmissions

38.3 μs at 160, at a utilization of 0.76–0.85. At low load fixed background work, such as heartbeats, raises the CPU per operation (285 μs at one client), so the comparison holds only near saturation.

Two vCPUs and the 50% curve. Two vCPUs act like one serial server of twice the speed, so $\sigma = \lambda D_{\mathrm{CPU}}/2$. At 50% writes the match is looser. The noisier 80-client runs give about 60 μs per operation against 52 μs from the fit, and their CPU cap of 33,000 ops/s is below the fit's asymptote of 38,200; the curve itself reaches 33,418 at 160 clients. From 30 to 50% writes the CPU per operation grows about 1.5 times and the single-client time $1/\lambda$ about 1.46 times, so σ hardly changes, as the bootstrap found.

Mitigations. To relieve the leader's CPU: a larger machine for it, `leaderServes=no` [6], observers for reads, and multi-operations. A faster log disk would not help, and turning `fsync` off costs throughput at high load as well as durability.

8 Conclusion

Within the levels we tested, what clients do matters more than how the servers are configured: the share of writes and the load explain most of the throughput variation, the log disk none, and `fsync` only through its interaction with load. With writes the scalability curves show contention but no crosstalk, and the contention does not change between 30 and 50% writes. Near saturation, the leader's CPU explains the bottleneck the fit implies.

Limitations. We measured scalability against load only, with three servers, and the number of servers appears only as a two-level factor. Several of our explanations also rest on measurements we did not take. We did not record the leader's share of the sessions in the factorial runs, probe the followers, measure off-CPU time per thread, trace the ext4 journal or check the 20 s windows for steady state. Treating two hyperthreads as two servers also overstates the leader's spare capacity. Finally, the workload has no creates or watches.

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A Stage I experiments not repeated in Stage II

These runs were made on the Stage I laptop with the servers in Docker containers, one run per cell and no confidence intervals. They come from a different testbed and are kept only as context.

YCSB. We ran YCSB [9] with its ZooKeeper binding on 3 servers (Table 8). At 16 threads the read-heavy workloads B and C reach about 6.7 times the throughput of the 50/50 workload A, and in A an update takes 3–4.5 times as long as a read on average (3,713 against 1,078 μs). B and C peak at 16 threads. Workload A jumps from 1,723 to 5,964 ops/s between 16 and 32 threads; with a single run per cell we cannot tell whether this is a real effect or noise, warm-up or garbage collection.

Table 8. YCSB throughput (ops/s), Stage I laptop, 3 servers, one run per cell.

Workload	1 thread	4	16	32
A (50/50 read/update)	790	1316	1723	5964
B (95/5)	3544	6927	11436	8080
C (100% read)	4538	9074	11913	8900

Leader failover. During a continuous benchmark on 3 servers we found the leader with `stat` and killed its container. The two survivors kept a majority and continued, and ZooKeeper logged an election of 211 ms (216 and 235 ms in a repetition). This is ZooKeeper's measurement of the election only, not the outage seen by clients.

Network delay. We added 20±2 ms to the egress of every server container with Pumba [10] (which drives `tc netem` inside each container's network namespace) and repeated the sweep with the quiet client (Table 9). Throughput per client stays at about 21.5 ops/s, one operation per 46.5 ms, at every load: the clients wait on the delay and the servers never saturate within 80 clients. The median read took 22.7 ms (one delayed hop) and the median write 106 ms (about five: forward, proposal, acknowledgment, commit, reply), as Fig. 1 predicts for a client attached to a follower.

Table 9. Throughput (ops/s) with and without a 20 ms egress delay, Stage I laptop, 3 servers, one run per cell.

Clients	1	5	10	20	40	80
No delay	2386	7016	11680	14832	16238	16255
20 ms delay	20.6	108.9	216.7	434.6	865.5	1709

B All effects of the factorial experiment

Table 10. All 31 effects ($\log_{10}$ model): q, the factor 10^{2q} from the low to the high level and the share of the variation (%). †: 95% interval includes zero.

		throughput			write p99		
Effect	Alias	q	×	%	q	×	%
A	BCDEF	+0.029	1.14	1.0	+0.021†	1.10	0.2
B	ACDEF	-0.228	0.35	60.3	+0.041	1.21	0.6
C	ABDEF	+0.149	1.98	25.6	+0.534	11.68	93.7
D	ABCEF	+0.015†	1.07	0.3	-0.042	0.82	0.6
E	ABCDF	-0.066	0.74	5.0	+0.061	1.32	1.2
F	ABCDE	-0.007†	0.97	0.1	+0.004†	1.02	0.0
AB	CDEF	-0.000†	1.00	0.0	-0.034	0.85	0.4
AC	BDEF	+0.018	1.09	0.4	+0.010†	1.05	0.0
AD	BCEF	+0.006†	1.03	0.0	+0.004†	1.02	0.0
AE	BCDF	-0.008†	0.97	0.1	+0.008†	1.04	0.0
AF	BCDE	+0.000†	1.00	0.0	+0.000†	1.00	0.0
BC	ADEF	-0.016	0.93	0.3	-0.020†	0.91	0.1
BD	ACEF	+0.009†	1.04	0.1	+0.015†	1.07	0.1
BE	ACDF	+0.003†	1.01	0.0	-0.022†	0.90	0.2
BF	ACDE	-0.006†	0.97	0.0	-0.003†	0.99	0.0
CD	ABEF	-0.056	0.77	3.6	+0.059	1.31	1.1
CE	ABDF	+0.007†	1.03	0.1	-0.013†	0.94	0.1
CF	ABDE	-0.007†	0.97	0.1	+0.010†	1.05	0.0
DE	ABCF	+0.008†	1.04	0.1	-0.002†	0.99	0.0
DF	ABCE	+0.002†	1.01	0.0	-0.006†	0.97	0.0
EF	ABCD	+0.021	1.10	0.5	-0.004†	0.98	0.0
ABC	DEF	+0.002†	1.01	0.0	-0.016†	0.93	0.1
ABD	CEF	+0.007†	1.03	0.1	+0.005†	1.03	0.0
ABE	CDF	+0.001†	1.00	0.0	-0.004†	0.98	0.0
ABF	CDE	-0.011†	0.95	0.1	+0.001†	1.00	0.0
ACD	BEF	+0.005†	1.02	0.0	-0.013†	0.94	0.1
ACE	BDF	-0.012†	0.95	0.2	+0.016†	1.07	0.1
ACF	BDE	+0.003†	1.01	0.0	+0.004†	1.02	0.0
ADE	BCF	-0.003†	0.98	0.0	-0.004†	0.98	0.0
ADF	BCE	+0.004†	1.02	0.0	-0.004†	0.98	0.0
AEF	BCD	-0.002†	0.99	0.0	-0.020†	0.91	0.1

C Probes at 80 clients: forceSync and log disk

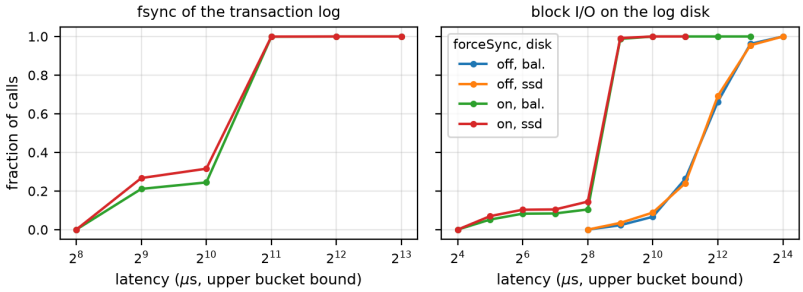


Fig. 4. CDFs of the `fsync` latency and of the log disk's block-I/O latency on the leader, from the $\log_2$ histograms of the probes (one point per bucket).

Table 11. Probe runs at 80 clients and 50% writes: throughput, ZooKeeper's counters and the leader's CPU per thread.

`forceSync`, log disk	off, bal.	off, ssd	on, bal.	on, ssd
ops/s with probes	29276	28739	26622	26674
ops/s without (mean of 2)	28795	24992	28731	27301
fsync calls (eBPF)	0	0	21393	23120
fsync mean, eBPF (μs)	–	–	1064	986
fsync mean, ZooKeeper (ms)	–	–	0.77	0.70
requests per flush	1.9	1.8	17.4	16.0
to quorum ACK (ms)	1.14	1.16	2.56	2.52
CPU of the leader (cores of 2)				
replication to followers	0.67	0.67	0.55	0.59
client connections (I/O)	0.40	0.38	0.40	0.39
request preparation	0.20	0.21	0.25	0.25
commit	0.19	0.18	0.17	0.16
log sync	0.10	0.09	0.10	0.11
commit workers	0.06	0.05	0.06	0.06
whole JVM	1.73	1.70	1.61	1.62

D Response time model

Table 12. Mean response time from the Amdahl fit and Little's law, against measured latencies (all operations).

Clients N	1	8	32	64	128	160
$R(N)$ predicted (ms)	0.48	0.61	1.06	1.66	2.85	3.45
N/X measured (ms)	0.46	0.61	1.05	1.66	2.81	3.45
median (ms)	0.22	0.28	0.49	0.87	1.64	2.42
95th perc. (ms)	1.15	1.79	2.95	4.48	7.50	8.75

E Per-stage times at every load level

Table 13. Leader per-stage times (ms, mean of the replicates) and requests per `fsync`; the write latency is pooled over all servers. 1 ms timer resolution.

Leader, mean per run	$N = 1$	$N = 16$	$N = 64$	$N = 160$
prep queue (zxid) (ms)	0.02	0.05	0.35	1.37
sync queue (ms)	0.03	0.43	0.66	0.76
sync queue + flush (ms)	0.55	1.08	1.86	2.14
fsync (ms)	0.06	0.22	0.69	0.75
until quorum ACK (ms)	0.72	1.34	2.26	2.79
commit to applied (ms)	0.02	0.08	0.50	1.13
server-side write latency (ms)	0.87	1.61	3.26	5.57
server-side read latency (leader) (ms)	0.05	0.12	0.90	2.79
requests per fsync	1.0	4.1	13.0	17.6

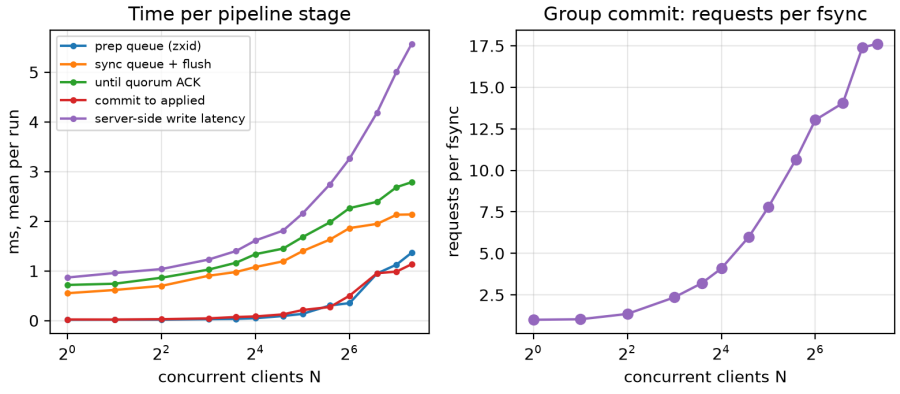


Fig. 5. Leader per-stage times (left) and requests per `fsync` (right) at every level of the USL curve.

F Model checks

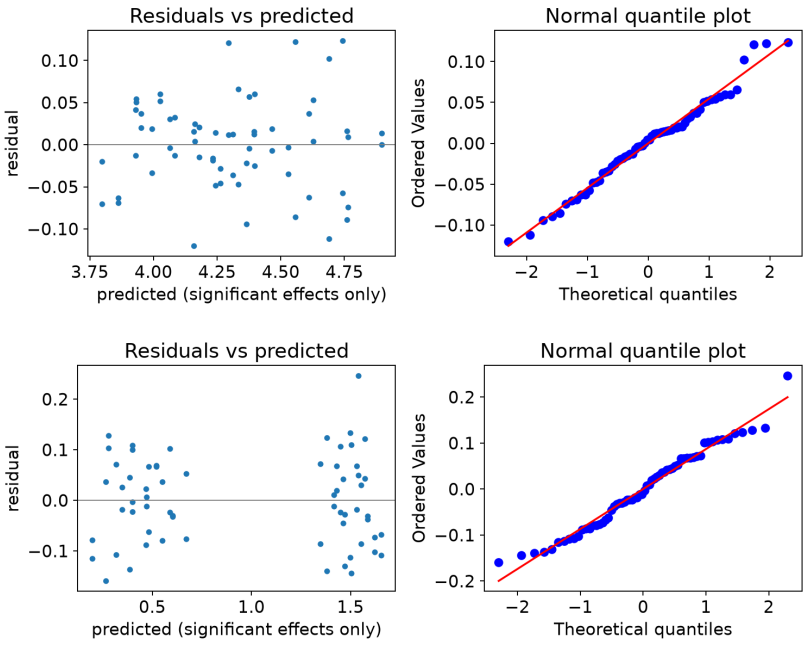


Fig. 6. Residuals of the $\log_{10}$ effects model against the predicted value (left) and normal quantile plots (right): throughput (top) and write p99 (bottom).